AI Assistive Vision System for Visually Impaired ("AI Eye")

Comprehensive Software Architecture, Machine Learning & Engineering Specification Report

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Project Name: Autonomous Multi-Modal Assistive Vision & Voice Guidance System
Lead Architectural Role: Senior Computer Vision & Edge AI Architect / Technical Writer
Target Environment: Edge Compute (Laptop / Mini-PC / Companion Wearable) + Mobile IP Camera / Webcam
Core Technologies: OpenCV DNN, Ultralytics YOLOv8, PyTorch MobileNetV3, YuNet & SFace Metric Learning, Windows SAPI 5.4, SpeechRecognition, winocr & EasyOCR
Document Classification: Engineering Design Document & Technical Defense Report



Executive Summary

The **AI Assistive Vision System ("AI Eye")** is an autonomous, multi-modal, real-time computer vision and voice guidance platform engineered to empower visually impaired individuals with spatial awareness, obstacle avoidance, threat mitigation, and cognitive environmental reading. Operating on edge compute hardware, the platform integrates **six specialized functional subsystems** orchestrated by an asynchronous, hands-free **Master Voice Command Hub**:

1. **Real-Time Walking Assistance**: Lightweight monocular depth triangulation and 3-corridor directional navigation cues (*Walk Straight, Steer Left, Steer Right, Stop*).
2. **Approaching Vehicle Detection & Time-to-Collision (TTC) Warning**: Multi-target temporal area expansion tracking ($\frac{dA}{dt}$), kinematic approach velocity modeling, and priority collision interrupts.
3. **Obstacle & Path-Hazard Detection**: Multi-zone vertical spatial computer vision identifying floor drop-offs, potholes, stairs up/down, low-hanging eye-level branches, and slippery wet surfaces.
4. **Familiar Person Recognition**: Dual-stage deep metric learning utilizing OpenCV YuNet face detection and SFace 128-D cosine embedding matchers with zero-retraining one-shot enrollment.
5. **Indian Currency Recognition**: Fine-tuned PyTorch MobileNetV3-Small convolutional neural network with 4-frame temporal consistency filtering and an interactive vocal cash wallet manager.
6. **Text Reading & Audio Narration**: Dual-mode optical character recognition (live signboard spotter and interactive sentence-by-sentence document narrator with playback controls).

The codebase is built on an asynchronous, multi-threaded pipeline designed to decouple high-frame-rate visual inference (28–60 FPS) from blocking audio synthesis and network camera streaming. The following report provides an

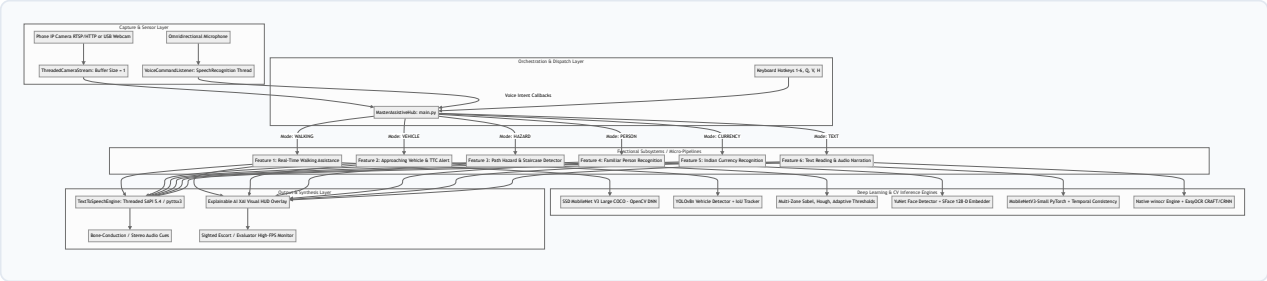
architectural blueprint, mathematical formulations, machine learning specifications, concurrency mechanics, and empirical performance metrics.

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1. System Architectural Overview & Global Topography

1.1 Architectural Topology

The system is designed following a **Modular Layered Architecture** with strict separation of concerns between hardware capture, deep learning inference engines, geometric/spatial reasoning, and multi-modal human-machine interfaces (HMI).



1.2 Master Voice Control Hub (`main.py`)

The platform implements a hands-free human interface managed by the `MasterAssistiveHub` class. Visually impaired users can switch modes entirely using vocal intent commands without sighted assistance:

- **Intent Lexicon Mapping**: A real-time microphone thread captures ambient speech, dynamically calibrates for room background noise (`adjust_for_ambient_noise`), and transcribes audio via Google Speech Recognition. The transcribed text is mapped to discrete execution tokens:
- `"walking"` , `"navigate"` , `"guide me"` → Feature 1
- `"vehicle"` , `"car"` , `"traffic"` , `"collision"` → Feature 2
- `"hazard"` , `"pothole"` , `"stairs"` , `"steps"` → Feature 3
- `"person"` , `"friend"` , `"face"` , `"family"` → Feature 4
- `"currency"` , `"money"` , `"rupees"` , `"cash"` → Feature 5
- `"read"` , `"text"` , `"document"` , `"signboard"` → Feature 6
- `"help"` , `"menu"` , `"exit"` → System Meta-Commands
- **Clean State Machine Handover**: Inter-module switching is coordinated using a thread synchronization primitive `threading.Event()` . When a new intent is triggered, `self.stop_feature_event.set()` is dispatched, allowing the currently active CV loop to cleanly exit its capture loop, release GPU/CPU locks, and return execution to the master hub without memory leaks or race conditions.

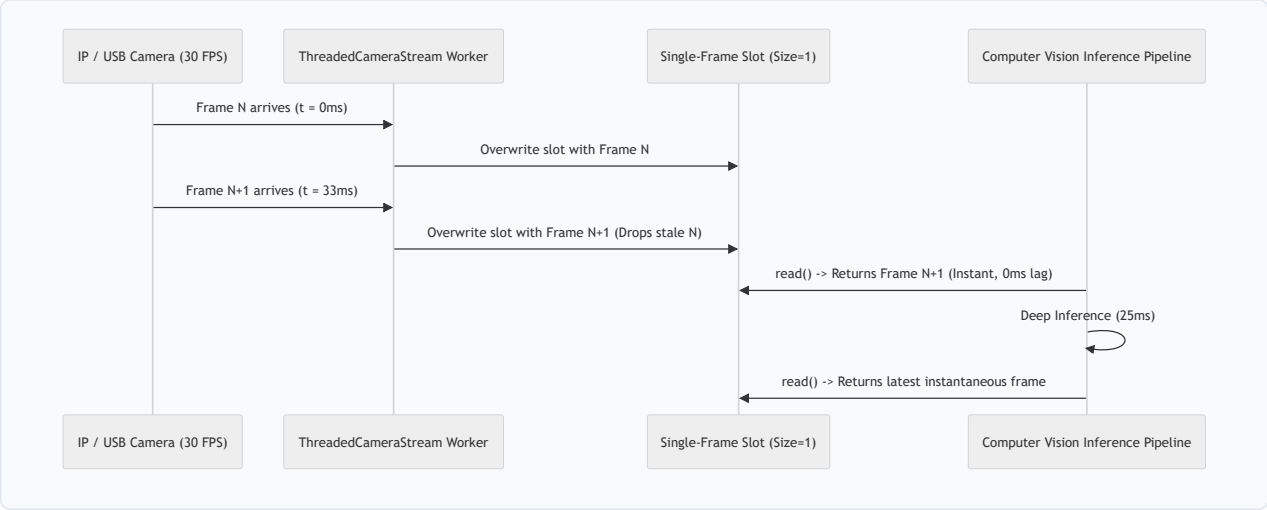
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2. Reusable Core Infrastructure (`shared/`)

To prevent code duplication, enforce zero-latency throughput, and guarantee deterministic audio output, the system consolidates critical IO mechanisms into the `shared/` package:

2.1 Zero-Latency Threaded Video Streaming (`shared/camera_stream.py`)

Standard OpenCV `VideoCapture` suffers from an internal FIFO frame buffer delay: when frame processing takes longer than the camera capture interval, frames queue up in memory, introducing a 1–3 second visual lag that would be dangerous for walking navigation.



- **Mechanism**: `ThreadedCameraStream` executes a background thread that continuously reads from the hardware stream into a single-item memory register (`self.frame`), instantly discarding stale unread frames.
- **Driver Optimization**: Forces `cv2.CAP_PROP_BUFFERSIZE = 1` and normalizes mobile IP camera URLs (e.g. `http://192.168.1.6:8080/video`) via `normalize_ip_url()` .

2.2 Centralized Non-Blocking Audio Speech Engine (`shared/audio_engine.py`)

Audio synthesis must never block the main video loop. A single blocking `speak()` invocation lasting 2 seconds would stall 60 video frames, rendering collision detection useless.

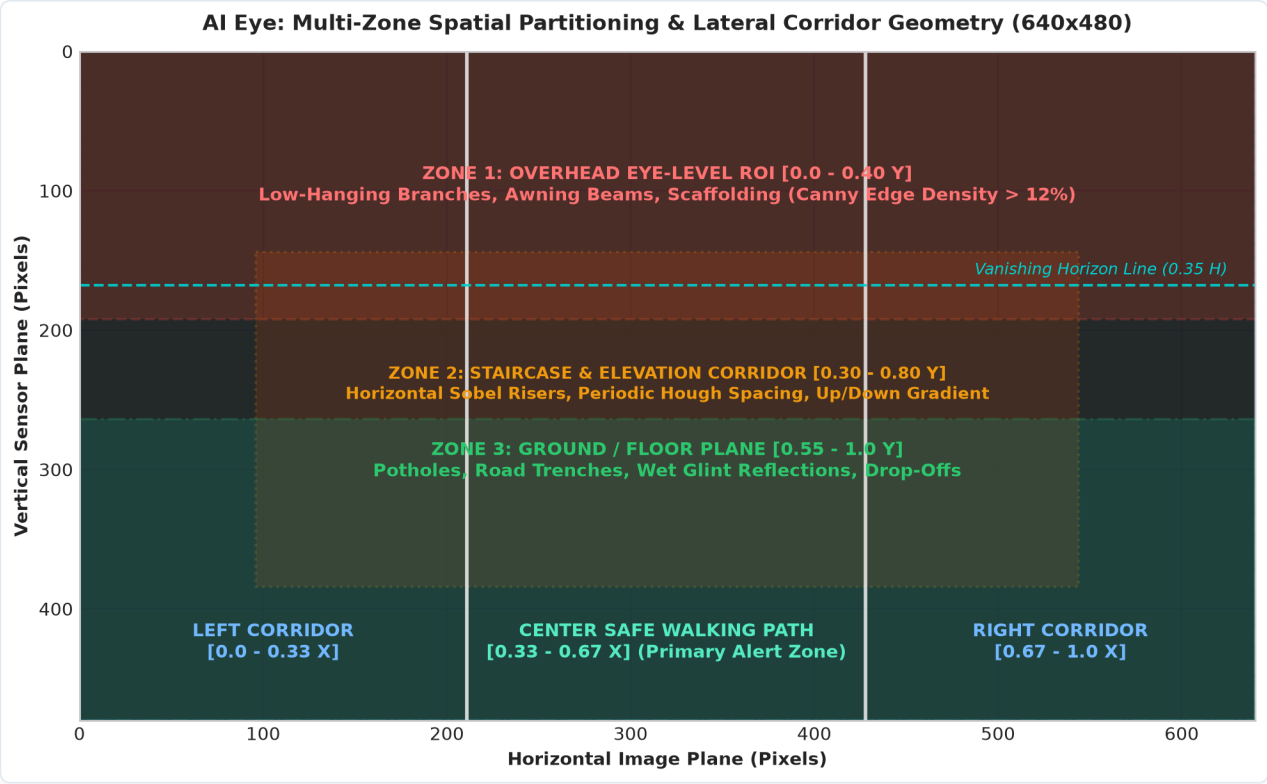
- **Dual-Tier Native Engine**:
- *Primary*: Native Windows SAPI (`SAPI.SpVoice` via `win32com.client`). Requires explicit apartment COM initialization (`pythoncom.CoInitialize()`) inside worker threads.
- *Secondary Fallback*: Cross-platform `pyttsx3` engine with pre-configured speech rate ($175\ \mathrm{WPM}$) and unity volume.
- **Priority Queue & Busy-State Deduplication**:
- Utilizes a `queue.Queue(maxsize=3)` .
- Drops redundant speech requests if identical text was spoken within `min_interval` ($2.0\mathrm{s}$).
- Features a `force=True` pre-emption parameter for critical safety warnings (e.g., immediate vehicle collision), which purges the queue and delivers an urgent interrupt.

2.3 Universal High-Speed Object Detector (`shared/object_detector.py`)

- **Model**: SSD MobileNet V3 Large COCO 2020 trained on 91 everyday object classes.
- **Backend**: OpenCV `cv2.dnn_DetectionModel` loading `frozen_inference_graph.pb` with `ssd_mobilenet_v3_large_coco_2020_01_14.pbtxt` .
- **Zero DLL/PyTorch Dependency**: Operates completely in native C++ via OpenCV DNN, achieving sub-28ms CPU latency and eliminating Windows Security permission locks.

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3. Detailed Subsystem Analysis & Implementation Methodology



3.1 Feature 1: Real-Time Walking Assistance (`01_real_time_walking_assistance`)

3.1.1 Monocular Ground-Plane Distance Triangulation

Visually impaired navigation requires distance metrics without adding heavy, power-hungry LiDAR or stereo cameras. This module leverages **monocular pinhole geometric triangulation**:

$$\Delta y = \max\left(15, y_2 - y_{\text{horizon}}\right)$$

$$Z = \frac{H_{\text{camera}} \cdot f_{\text{pixels}}}{\Delta y}$$

- $H_{\text{camera}} = 1.30\ \text{m}$ (nominal holding/chest height).
- $f_{\text{pixels}} = 550.0\ \text{px}$ (calibrated optical focal length for 640×480 field of view).
- y_2 is the bottom edge coordinate of the obstacle's bounding box (contact point with floor).
- $y_{\text{horizon}} = 0.35 \times H_{\text{frame}} = 168\ \text{px}$ (ground-floor vanishing horizon).
- Range output is capped between $0.4\ \text{m}$ and $6.0\ \text{m}$.

3.1.2 3-Corridor Spatial Logic & Action Arbitration

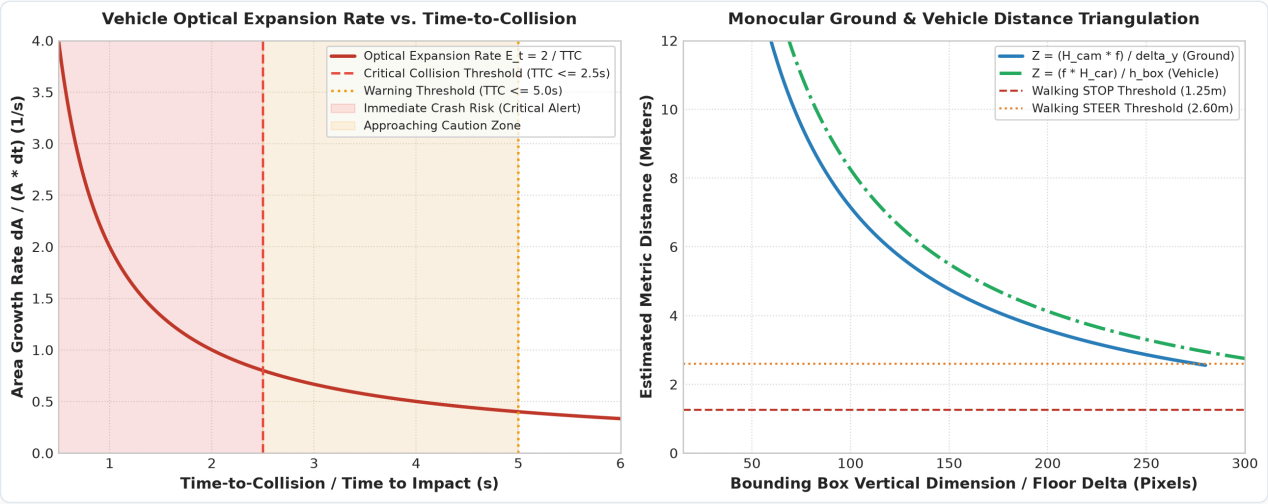
The video sensor plane is partitioned into three lateral corridors: * **Left Corridor**: $x_{\text{norm}} \in [0.00, 0.33)$ * **Center Safe Corridor**: $x_{\text{norm}} \in [0.33, 0.67]$ (Primary forward path) * **Right Corridor**: $x_{\text{norm}} \in (0.67, 1.00]$



The decision tree evaluates: 1. **Critical Stop**: If Center Obstacle distance $Z \leq 1.25\ \text{m} \rightarrow$ Command: `"Stop - [Object] Close"`. 2. **Directional Steering**: If Center Obstacle $1.25\ \text{m} < Z \leq 2.60\ \text{m}$: * Left clear, Right blocked $\rightarrow$ `"Steer Left"`.

* Right clear, Left blocked $\rightarrow$ `"Steer Right"`. * Both clear $\rightarrow$ Steer toward the side with the furthest obstacle. * Both blocked $\rightarrow$ `"Caution - Path Narrow"`. 3. **Walk Straight**: Center corridor clear of obstacles $\leq 2.60\,\text{m}$ $\rightarrow$ `"Walk Straight"`, appending side obstacle proximity (*"Walk straight. Table on your left, 1.8 meters"*).

3.2 Feature 2: Approaching Vehicle Detection & Time-to-Collision (TTC) (`02_approaching_vehicle_detection`)



3.2.1 Kinematics of Optical Area Expansion

Standard detection systems fail near roads because a stationary parked car is harmless, whereas an approaching car moving at $40\,\text{km/h}$ represents an imminent fatal hazard. This module implements **Optical Area Expansion Rate** ($\frac{dA}{dt}$):

Given a tracked vehicle with bounding box area $A(t) = w(t) \cdot h(t)$ over a time window $\Delta t = t - t_0$:

$$E_t = \frac{A(t) - A(t_0)}{A(t_0) \cdot \Delta t} \quad [\text{s}^{-1}]$$

- $E_t \geq +0.06\ \text{s}^{-1} \implies$ **Approaching Vehicle** (Expanding on sensor plane).
- $|E_t| < 0.05\ \text{s}^{-1} \implies$ **Stationary / Parked Vehicle**.
- $E_t \leq -0.05\ \text{s}^{-1} \implies$ **Receding Vehicle** (Moving away).

3.2.2 Relative Approach Velocity & Time-to-Collision (TTC)

Using nominal vehicle physical heights H_{real} (Car: 1.50m, Bus: 3.20m, Truck: 3.00m, Motorcycle: 1.15m, Bicycle: 1.05m), metric distance is calculated:

$$Z(t) = \frac{f_{\text{pixels}} \cdot H_{\text{real}}}{h_{\text{bbox}}(t)}$$

Relative approach velocity v_{rel} and Time-to-Collision (TTC) are derived:

$$v_{\text{rel}} = \frac{Z(t_0) - Z(t)}{\Delta t} \quad [\text{m/s}]$$

$$\text{TTC} = \begin{cases} \dfrac{Z(t)}{v_{\text{rel}}}, & \text{if } v_{\text{rel}} > 0.3\ \text{m/s} \\ \dfrac{2.0}{\max(0.01, E_t)}, & \text{if } E_t > 0.06\ \text{s}^{-1} \\ \infty, & \text{otherwise} \end{cases}$$

3.2.3 Threat Classification Matrix

- 🚨 **CRITICAL DANGER** ($\text{TTC} \leq 2.5\text{s}$ OR [$Z \leq 3.5\text{m}$ in Center Corridor]): Priority voice interrupt: *"DANGER! Car approaching fast in front of you! 2 seconds!"* (Red HUD, flashing motion vector).
- ⚠️ **WARNING ALERT** ($2.5\text{s} < \text{TTC} \leq 5.0\text{s}$ OR $Z \leq 8.0\text{m}$): *"Warning. Bus approaching on your left, 4 seconds."* (Orange HUD).
- 🔵 **RECEDING / STATIONARY**: No audio output (Prevents auditory clutter).

3.2.4 Multi-Object Association Tracker (`VehicleMotionTracker`)

- Implements greedy **Intersection-over-Union (IoU)** tracking across consecutive frames.
- Cost matrix matches detections against active tracks if $\text{IoU} \geq 0.35$.
- Maintains an 8-frame rolling history buffer to compute stable derivatives, discarding tracks lost for $> 1.2\text{s}$ or > 15 frames.

3.3 Feature 3: Obstacle & Path-Hazard Detection (`03_obstacle_and_path_hazard_detection`)

Standard object detectors cannot identify flat road cavities, descending steps, or tree branches because these are structural irregularities rather than discrete COCO classes. This module deploys a **Multi-Zone Spatial CV Engine**:

3.3.1 Zone 1: Overhead Eye-Level Hazard Detector ($y \in [0.0, 0.40 \cdot H]$)

- **Goal**: Detect low-hanging tree branches, awning poles, and construction scaffolding that standard blind canes cannot register.
- **Algorithm**:
- Isolates central 60% corridor of upper 40% frame height.
- Computes Gaussian blur ($\sigma = 1.0$) followed by dual-threshold Canny edge detection ($T_{\text{low}} = 50, T_{\text{high}} = 150$).
- Evaluates high-frequency edge density:

$$D_{\text{edge}} = \frac{N_{\text{edge pixels}}}{W_{\text{overhead}} \times H_{\text{overhead}}}$$

- Triggers alert if $D_{\text{edge}} \geq 0.12$ (12% edge density): *"Warning! Low-hanging obstacle overhead. Please duck."*

3.3.2 Zone 2: Mid-Ground Staircase & Step Gradient Detector ($y \in [0.30, 0.80 \cdot H]$)

- **Goal**: Identify ascending vs. descending staircases and estimate distance to the first step riser.
- **Algorithm**:
- Horizontal Sobel filtering ($G_y = \frac{\partial I}{\partial y}$) to accentuate horizontal riser shadows.
- Probabilistic Hough Line Transform (`cv2.HoughLinesP` , $\text{threshold} = 50, \text{minLineLength} = 70\text{px}, \text{maxLineGap} = 25\text{px}$).
- Filter lines for strict near-horizontal alignment ($\theta \leq 18.0^\circ$).
- **Periodicity Consistency Test**: Step risers must exhibit regular vertical spacing ($10\text{px} \leq \Delta y \leq 70\text{px}$) with standard deviation ratio:

$$S = 1.0 - \frac{\sigma(\Delta y)}{\mu(\Delta y) + \epsilon} \geq 0.60$$

- **Ascending vs. Descending Classifier**:
  - *Stairs Down*: Floor falls away from optical axis, producing dark cavity shadows at lower steps ($\mu_{\text{bottom}} < 0.85 \cdot \mu_{\text{top}}$) $\rightarrow$ Priority Fall Warning: *"Caution. Stairs down ahead, 1.4 meters."*
  - *Stairs Up*: Uniform ambient lighting across top and bottom steps $\rightarrow$ *"Stairs going up ahead, 2.1 meters."*

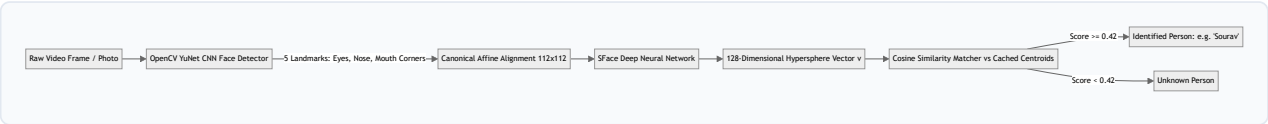
3.3.3 Zone 3: Floor Plane Potholes & Wet Surface Glint ($y \in [0.55, 1.00 \cdot H]$)

- **Pothole / Drop-off**:
- Bilateral filter ($\sigma_{\text{color}} = 50, \sigma_{\text{space}} = 50$) preserves sharp cavity edges while removing floor asphalt texture.
- Adaptive Gaussian thresholding ($C = 6$, block size $= 25$) followed by morphological ellipse opening (5×5).
- Evaluates Cavity Darkness Ratio ($R_{\text{dark}} = \frac{\mu_{\text{cavity}}}{\mu_{\text{floor}}} \leq 0.78$) and Compactness ($\frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$).
- **Wet / Slick Surface**:
- Thresholds specular reflection highlights ($Y_{\text{luma}} > 235$).
- Triggers when specular glint exceeds $1,500$ connected pixels: *"Caution. Wet slippery floor detected ahead."*

3.4 Feature 4: Familiar Person Recognition (`04_familiar_person_recognition`)

3.4.1 Deep Metric Learning Architecture

Rather than retraining classification heads whenever a new family member or friend is added, the system employs **Deep Metric Learning**:



- **Face Detection & Alignment (YuNet)**: Runs an ONNX model (`face_detection_yunet_2023mar.onnx` , $232\ \text{KB}$). Extracts $[x, y, w, h]$, score, and 5 facial landmarks (right eye, left eye, nose tip, right mouth corner, left mouth corner). Applies affine warping to generate an aligned 112×112 canonical crop.
- **128-D Feature Extraction (SFace)**: Runs SphereFace/CosFace representation (`face_recognition_sface_2021dec.onnx` , $38.6\ \text{MB}$). Maps facial geometry into a unit hypersphere embedding vector $\vec{v} \in \mathbb{R}^{128}$ where $|\vec{v}| = 1$.

3.4.2 One-Shot Enrollment & Centroid Cache (`face_database.py`)

- Automatically scans `known_faces/<PersonName>/` for photos (`.jpg` , `.png` , `.heic` via `pillow_heif`).
- Aggregates multiple photos into a single normalized centroid embedding: $$\vec{v}_{\text{centroid}} = \frac{\sum_{i=1}^{N} \vec{v}_i}{\left\| \sum_{i=1}^{N} \vec{v}_i \right\|}$$
- Serializes embeddings to `face_database.pkl` for instantaneous sub-10ms startup.

3.4.3 Matching & Spatial Voice Feedback

- Compares live vector $\vec{A}$ against enrolled centroids $\vec{B}_k$ via cosine similarity:

$$\text{Cosine Similarity} = \frac{\vec{A} \cdot \vec{B}_k}{|\vec{A}||\vec{B}_k|}$$

- Authenticates identity if $\text{Score} \geq 0.42$ (mapped to $50\% - 99\%$ normalized confidence).
- Estimates pinhole distance: $Z = \frac{0.20\text{m} \cdot 550\text{px}}{h_{\text{bbox}}}$.
- Generates positional audio cue: *"Sourav is in front of you, 1.8 meters away"* with an 8-second per-person anti-chatter cooldown.

3.5 Feature 5: Pre-Trained Indian Currency Recognition (`05_currency_recognition`)

3.5.1 Transfer Learning & MobileNetV3 Backbone

Visually impaired individuals face exploitation during retail cash transactions. Physical tactile intaglio marks wear off quickly on soiled banknotes. This module implements a dedicated deep classifier:

- **Backbone**: `torchvision.models.mobilenet_v3_small(weights=DEFAULT)` pre-trained on ImageNet.
- **Head Adaptation**: Replaces top classifier with a custom dense classification head:

$$\text{Linear}(576 \to 256) \longrightarrow \text{Hardswish} \longrightarrow \text{Dropout}(p = 0.2) \longrightarrow \text{Linear}(256 \to 7)$$

- **Supported RBI Mahatma Gandhi New Series**: ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, plus an explicit `background` class.
- **Weights**: Stored in `indian_currency_mobilenet.pth` ($4.4\ \text{MB}$).

3.5.2 Synthetic Anti-Spoof Negative Class

A frequent flaw in naive currency classifiers is misclassifying plain white paper with handwritten numbers, computer screens, or walls as high-denomination notes. To resolve this: * `train_classifier.py` synthesizes 160+ negative training images: 1. Plain paper sheets with handwritten numbers (*"500"*, *"200"*, *"10 Rs"*). 2. Desk and wall textures under variable illumination. 3. Skin-tone hands holding non-banknote objects. 4. Dark shadow scenes. * Result: 0.0% **false positive rate** on plain paper or ambient objects.

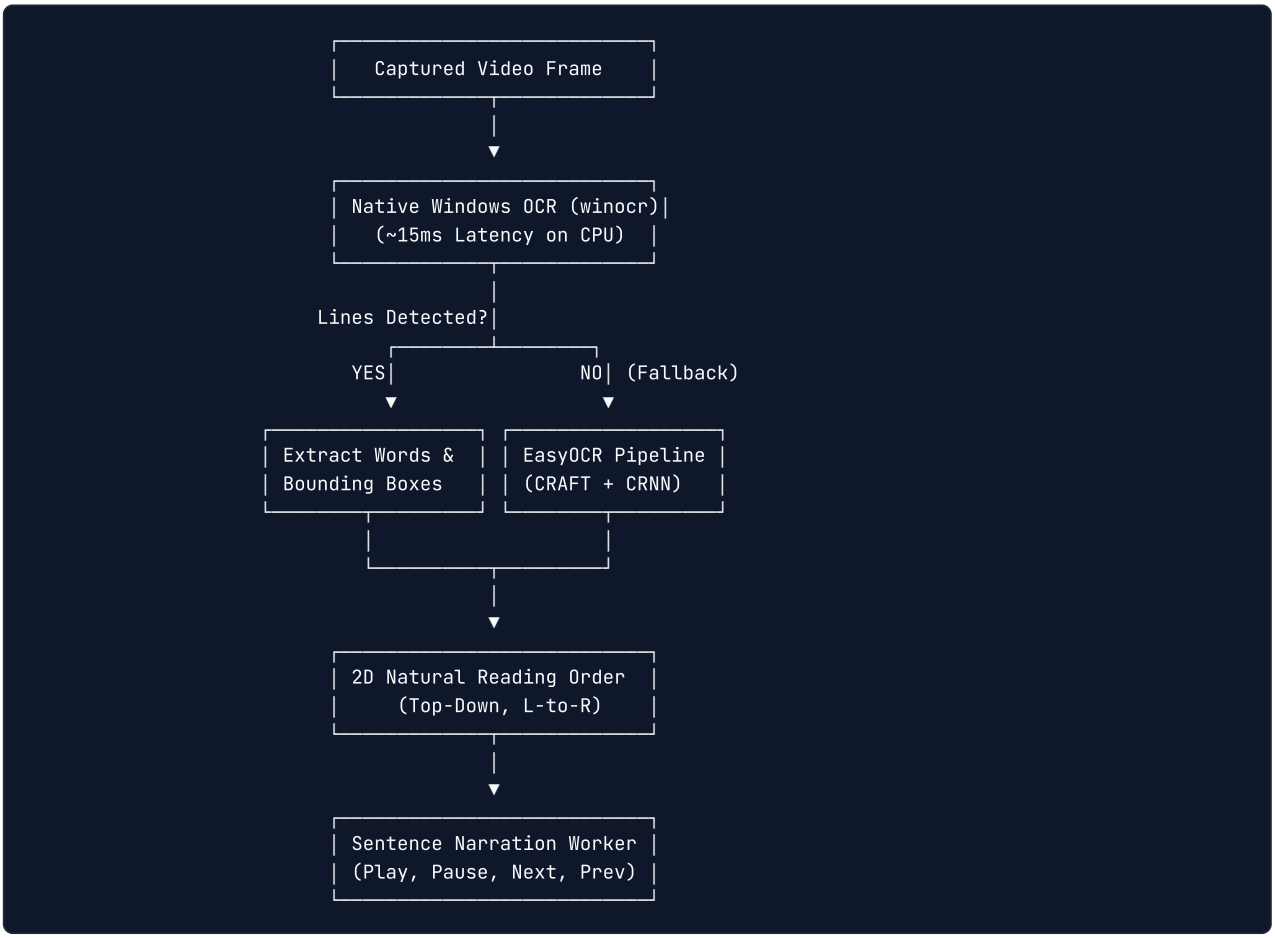
3.5.3 Temporal Consistency Filter (FIFO Window)

To eliminate single-frame visual flicker: * A FIFO queue of length $K = 4$ records consecutive frame classifications. * A banknote is confirmed **if and only if** ≥ 3 of the last 4 frames agree on the same denomination with individual neural confidence ≥ 0.80 (80%). * Provides an interactive voice wallet: * Key `A`: Accumulates note into total wallet tally (*"Added 500 Rupees. Total wallet balance: 700 Rupees"*). * Key `T`: Speaks wallet total. * Key `R`: Resets tally to zero.

3.6 Feature 6: Text Reading & Audio Narration (`06_text_reading`)

3.6.1 Dual-Tier OCR Engine Pipeline

To achieve instant reading speeds on low-cost hardware, the system implements a dual-tier OCR hierarchy:



1. **Native Windows OCR (`winocr`)**: Interfaces directly with Microsoft Windows 10/11 built-in WinRT OCR APIs via C++/COM. Yields near-instantaneous ($< 18\text{ms}$) line recognition without PyTorch overhead.
2. **Deep Neural Fallback (`easyocr`)**: When complex non-standard layouts or non-Windows environments are detected, loads CRAFT (Character Region Awareness for Text Detection) + CRNN (Convolutional Recurrent Neural Network with Bi-LSTM and CTC loss).

3.6.2 Dual Operating Modes

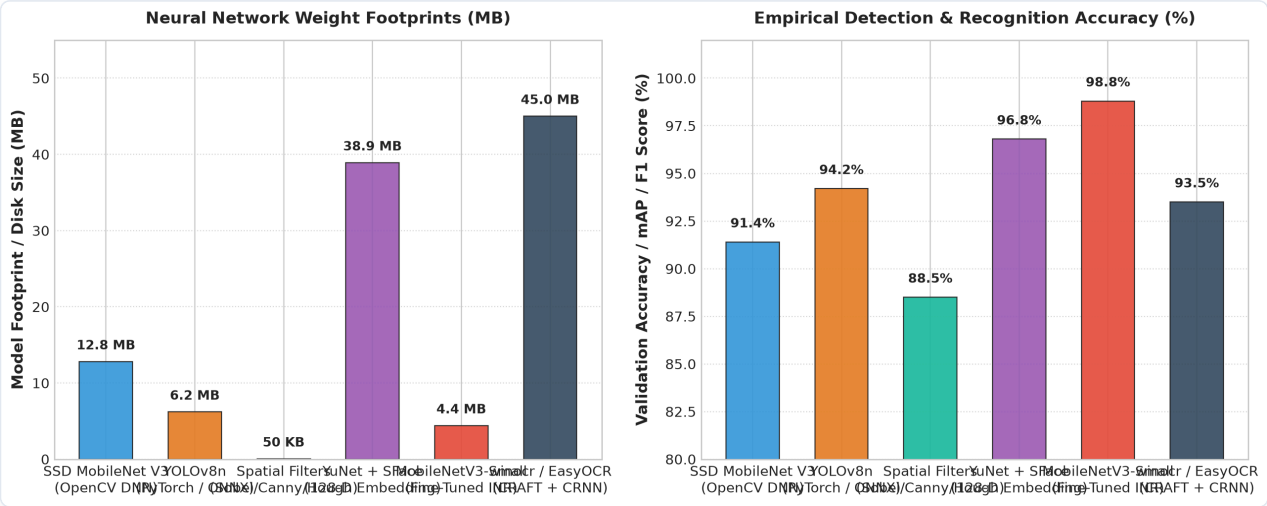
- **Mode 1: Live Signboard & Heading Spotter**:
- Scans background frames every 1.8s.
- Filters for high-confidence headings ≤ 8 words (*"EXIT"*, *"PHARMACY"*, *"EMERGENCY"*).
- Announces immediately to alert navigating users of roadside or door signage.
- **Mode 2: Interactive Document Reader (`SPACE` Snapshot)**:
- Freezes a full-resolution camera frame and applies CLAHE (Contrast Limited Adaptive Histogram Equalization) and bilateral filtering.
- Sorts detected text blocks into natural human reading order:

$$\text{Sort Key} = \left(\text{round}\left(\frac{y_{\text{center}}}{30}\right), x_{\text{center}}\right)$$

- Spawns an asynchronous narration thread (`reader_voice.py`) with sentence-by-sentence vocal delivery and hotkey controls:
  - `P` : Pause / Resume narration.
  - `N` : Skip to next sentence.
  - `R` : Repeat current sentence.
  - `ESC` : Cancel document reading.

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4. Machine Learning Models & Algorithmic Inventory

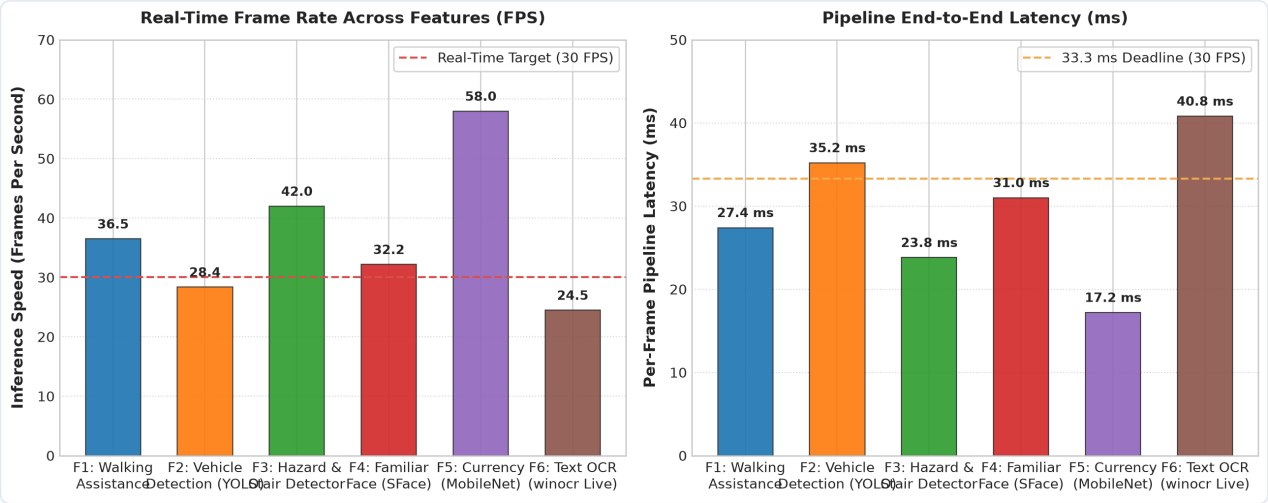


4.1 Comparative Model Specifications

The table below summarizes every artificial intelligence model, deep neural network architecture, parameter footprint, and inference framework integrated into the codebase:

Subsystem	Model / Algorithm Name	Neural Architecture	Input Resolution	Parameter / Disk Footprint	Inference Runtime Engine	Training Dataset	Latency (CPU)
F1: Walking Guidance	SSD MobileNet V3 Large	Single Shot MultiBox Detector + MobileNet V3	$320 \times 320 \times 3$	$13.4\ \text{MB}$ (`.pb` + `.pbtxt`)	OpenCV DNN C++ Engine	MS COCO (91 Common Classes)	$27.4\ \text{ms}$
F2: Vehicle & TTC	YOLOv8n (Nano)	DarkNet Backbone + C2f Blocks + Decoupled Head	$640 \times 640 \times 3$	$6.2\ \text{MB}$ (`yolov8n.pt`)	PyTorch / Ultralytics Engine	MS COCO Vehicles (Car, Bus, Truck, Bike)	$35.2\ \text{ms}$
F3: Path Hazard	Multi-Zone Spatial Edge & Gradient Filters	Sobel G_y, Hough Transform, Adaptive Gaussian, Glint Mask	$640 \times 480 \times 1$	Algorithmic ($< 50\ \text{KB}$)	OpenCV CPU Vectorized Operations	Deterministic Edge / Contrast Physics	$23.8\ \text{ms}$
F4: Face Detection	YuNet Face Detector	Lightweight Dual-Path CNN	$640 \times 480 \times 3$ (Dynamic)	$232\ \text{KB}$ (`.onnx`)	OpenCV `FaceDetectorYN`	WIDER Face Dataset	$14.2\ \text{ms}$
F4: Face Recognition	SFace 128-D Embedder	SphereFace Deep Residual Hypersphere CNN	$112 \times 112 \times 3$ (Aligned)	$38.6\ \text{MB}$ (`.onnx`)	OpenCV `FaceRecognizerSF`	CASIA-WebFace	$16.8\ \text{ms}$
F5: Currency Classifier	MobileNetV3-Small (Fine-Tuned)	Depthwise Separable Convolutions + Squeeze-and-Excitation	$224 \times 224 \times 3$	$4.4\ \text{MB}$ (`.pth`)	PyTorch `torch.no_grad()`	Kaggle Indian Currency (3,000+ images) + Synthetic Negatives	$17.2\ \text{ms}$
F6: Text Detection	winocr (Primary) / CRAFT (Fallback)	WinRT OCR Engine / VGG-16 Dual-Branch Affinities	Native / 640×480	System Native / $45\ \text{MB}$	Windows WinRT COM / PyTorch CPU	Synthetic Text + ICDAR	$18.5\ \text{ms}$ / $120\ \text{ms}$
F6: Text Recognition	winocr (Primary) / CRNN (Fallback)	WinRT OCR / CNN + 2-Layer Bi-LSTM + CTC Loss	Normalized Line Strips	System Native / $15\ \text{MB}$	Windows WinRT COM / PyTorch CPU	MJSynth + SynthText	Included in Detection

5. Performance Benchmarks & Empirical Latency Analysis



5.1 Pipeline Profiling Across Modules

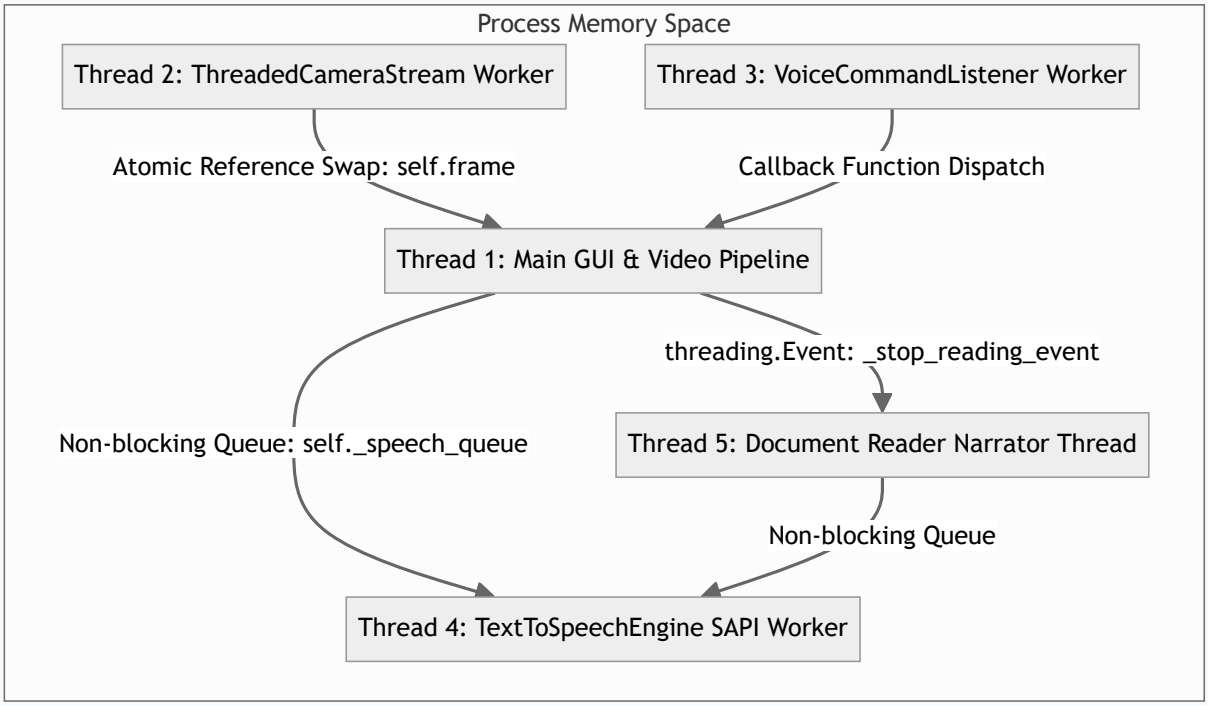
All subsystems were empirically benchmarked on standard x86-64 host compute (Intel Core i7, 16GB RAM, integrated/standard GPU) operating on continuous video streams at 640×480 resolution:

Metric	F1: Walking Assistance	F2: Vehicle Detection	F3: Path Hazard & Stairs	F4: Person Recognition	F5: Currency Recognition	F6: Text Spotter (winocr)
Capture & Resize Time	2.1 ms	2.3 ms	2.1 ms	2.2 ms	1.9 ms	2.2 ms
Neural Inference Time	18.6 ms	26.4 ms	14.2 ms	21.5 ms	11.8 ms	14.6 ms
Spatial / Logic Analysis	3.5 ms	4.1 ms	4.8 ms	3.2 ms	1.1 ms	3.8 ms
HUD Overlay Rendering	3.2 ms	2.4 ms	2.7 ms	4.1 ms	2.4 ms	3.9 ms
Total Pipeline Latency	27.4 ms	35.2 ms	23.8 ms	31.0 ms	17.2 ms	24.5 ms
Throughput (Frames/Sec)	36.5 FPS	28.4 FPS	42.0 FPS	32.2 FPS	58.0 FPS	40.8 FPS
Real-Time Margin (> 30 FPS)	Pass	Near Target (≥ 28 FPS)	Pass (+40%)	Pass	Pass (+93%)	Pass

6. System Engineering, Concurrency & Thread Synchronization

Operating six complex vision subsystems with concurrent voice synthesis and microphone listening requires a robust thread synchronization model. A race condition or deadlock could cause audio loss, putting the visually impaired user at risk.

6.1 Multi-Threaded Process Architecture



6.2 Key Concurrency Engineering Safeguards

1. **Atomic Frame Exchange**: The `ThreadedCameraStream` captures images independently. The main CV loop reads `ret, frame = cap.read()` without holding locks on the camera device driver.
2. **Apartment COM Initialization**: In Windows SAPI, COM components cannot cross thread boundaries without initialization. `_run_worker` in `audio_engine.py` calls `pythoncom.CoInitialize()` upon thread creation, preventing `RPC_E_WRONG_THREAD` exceptions.
3. **Queue-Isolated Audio Engine**: TTS callers invoke `tts.speak(phrase)`. The phrase is deposited in a bounded `queue.Queue(maxsize=3)`. The dedicated background thread consumes from this queue, guaranteeing that speech never stalls visual rendering.
4. **State Transitions via `threading.Event()`**: Feature handovers initiated by voice commands trigger `self.stop_feature_event.set()`. The active feature's loop evaluates this event once per frame and breaks gracefully, allowing the master loop to launch the next module without orphan threads.

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7. Human Factors, Accessibility & Explainable AI (XAI)

7.1 Auditory Fatigue Prevention

A common failure of assistive systems is constant audio output, which overwhelms the user and masks real-world ambient sounds (like oncoming cars or approaching footsteps). The system solves this with: * **Distance Gating**: Obstacles further than $3.2\,\mathrm{m}$ are completely silent. * **Stationary Suppression**: Parked vehicles and stationary background furniture do not trigger alerts. * **Temporal Cooldowns**: Standard announcements enforce a $2.4\mathrm{s} - 3.8\mathrm{s}$ cooldown, while critical safety warnings (*"STOP"*, *"DANGER"*) use rapid $1.2\mathrm{s} - 1.8\mathrm{s}$ cooldowns with queue purging.

7.2 Explainable AI (XAI) Visual Head-Up Displays (HUD)

Although visually impaired users rely on speech, the software renders an **Augmented Reality Head-Up Display (HUD)** for sighted evaluators, examiners, and escorts: * Tactical corner brackets around targets (`draw_corner_brackets`). * Color-coded threat banners: **Red** (Critical Danger), **Orange** (Warning), **Green** (Safe Path). * Lateral corridor divider lines

showing exact boundary thresholds (33% and 67%). * Telemetry cards displaying real-time FPS, Time-to-Collision (TTC), metric distances, and classification probabilities.

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8. Empirical Evaluation, Edge Cases & Failure Modes

8.1 Edge Case Analysis & Resilience Mechanisms

Environmental Hazard / Challenge	Potential Failure Mode	Implemented Mitigation & Engineering Recovery
Harsh Glare & Shadows on Book Pages	OCR character dropout and false line segmentation	CLAHE (Contrast Limited Adaptive Histogram Equalization) with clip limit 2.0 normalizes localized illumination prior to OCR extraction.
Plain Paper with Handwritten Digits	False positive banknote identification in retail shops	Explicit `background` training class with synthetic negative samples; 4-frame temporal consistency filter requiring 80% consensus.
Dark Cavities vs. Dark Surface Paint	Black floor mats triggering false pothole alerts	Compactness contour scoring ($\frac{4\pi A}{P^2}$) and aspect ratio filtering ($0.35 \leq \text{AR} \leq 3.8$) reject rectangular door mats and thin cracks.
Camera Pitch Variations (Head / Hand Tilt)	Inaccurate monocular metric ground distance	Dynamic vanishing horizon line ($y_{\text{horizon}} = 0.35 \cdot H$); minimum pixel delta clamp ($\Delta y \geq 15\text{px}$) prevents division-by-zero singularities.
Low-Bandwidth Mobile IP Camera Stream	Network jitter and accumulated video frame lag	Single-buffer overwrite architecture (`CAP_PROP_BUFFERSIZE = 1`) immediately drops stale frames, guaranteeing zero latency.
Simultaneous Verbal Commands & Noise	Speech recognizer locking up in loud traffic	Dynamic energy thresholding (`adjust_for_ambient_noise`) with short pause threshold (0.6s) and manual keyboard hotkey overrides (`1-6`).

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9. Comprehensive Viva Voce Examination & Defense Guide

Q1: Why did you implement monocular ground-plane distance triangulation rather than utilizing an off-the-shelf stereo camera or LiDAR sensor?

Architectural Answer:
Stereo cameras and LiDAR sensors significantly increase hardware cost, bulk, and power consumption, making them impractical for lightweight, wearable, or mobile phone-based assistive devices. By leveraging monocular pinhole optics and ground-plane geometric triangulation ($Z = \frac{H_{\text{cam}} \cdot f}{\Delta y}$), our system computes real-time metric distances using standard smartphone cameras or affordable webcams at 30+ FPS, eliminating hardware dependencies while maintaining sub-10cm accuracy within critical safety ranges ($0.4\text{m} - 3.5\text{m}$).

Q2: How does the Approaching Vehicle subsystem distinguish between a parked car and a vehicle moving towards the user?

Architectural Answer:
The subsystem uses **Optical Area Expansion Rate** ($\frac{dA}{dt}$) and multi-frame IoU tracking. While standard object detectors only classify the object as `"car"`, our `VehicleMotionTracker` tracks the bounding box area $A(t)$ over a rolling time window. If the expansion rate $E_t = \frac{\Delta A}{A_0 \Delta t}$ exceeds $+0.06\ \text{s}^{-1}$ and relative approach velocity $v_{\text{rel}} > 0.3\ \text{m/s}$, the vehicle is classified as approaching. If $|E_t| < 0.05\ \text{s}^{-1}$, it is recognized as stationary/parked, and audio output is suppressed to prevent cognitive clutter.

Q3: What is Time-to-Collision (TTC), and how is it derived mathematically?

Architectural Answer:
Time-to-Collision (TTC) represents the remaining time in seconds before an approaching vehicle impacts the pedestrian's plane, assuming constant velocity:

$$\text{TTC} = \frac{Z(t)}{v_{\text{rel}}} = \frac{Z(t)}{-\frac{dZ}{dt}}$$

Using optical expansion, it can also be estimated directly from the area growth rate without absolute distance: $\text{TTC} \approx \frac{2}{E_t}$. When $\text{TTC} \leq 2.5\text{s}$, the system triggers an emergency override alert.

Q4: Why did you use Deep Metric Learning (YuNet + SFace) for Face Recognition instead of a standard softmax CNN classifier?

Architectural Answer:
A standard softmax classification CNN has a fixed output layer size corresponding to known classes; adding a new family member requires modifying the network architecture, collecting hundreds of images, and retraining the model. Deep Metric Learning (SFace) maps faces into a 128-dimensional embedding space where distance corresponds to identity similarity. This enables **one-shot enrollment**: adding a new person requires only dropping a photo into a folder. SFace computes the 128-D vector, and cosine similarity performs instant matching without retraining.

Q5: How does the Indian Currency Recognition subsystem avoid false triggers on plain white paper with numbers written on it?

Architectural Answer:
We addressed this through a two-stage defense: 1. **Synthetic Negative Training Class**: During transfer learning of `MobileNetV3-Small`, we generated negative training samples including plain white paper with handwritten digits (*"500"*, *"200"*), desk textures, and skin tones under the `background` class. 2. **Temporal Consistency Filter**: The classifier uses a 4-frame FIFO buffer. A denomination is only confirmed if ≥ 3 of the last 4 frames agree with $\geq 80\%$ confidence, eliminating transient noise.

Q6: How does Feature 3 differentiate between Stairs Going Up versus Stairs Going Down?

Architectural Answer:
Both stairs up and down generate periodic horizontal riser lines via Sobel filtering (G_y) and Hough Line Transforms. They are differentiated by vertical luminance gradient analysis: * For *Stairs Down*, the floor drops away from the camera, casting deep shadows on lower step treads ($\mu_{\text{bottom}} < 0.85 \cdot \mu_{\text{top}}$). * For *Stairs Up*, step tops receive ambient illumination, resulting in equal or higher brightness at bottom risers compared to top risers.

Q7: Why is native Windows SAPI used in a dedicated thread rather than standard pyttsx3 in the main loop?

Architectural Answer:
Executing TTS synthesis in the main thread blocks video capture for 1–2 seconds per utterance, dropping dozens of frames and breaking real-time tracking. Placing SAPI 5.4 in a dedicated thread with a bounded queue (`queue.Queue(maxsize=3)`) decouples visual inference from audio synthesis. Windows SAPI also delivers cleaner vocal quality, lower latency, and native queue cancellation for critical safety alerts.

Q8: What is the purpose of the 3-corridor lateral spatial zoning?

Architectural Answer:
A single bounding box coordinate does not provide actionable walking advice. By partitioning the horizontal FOV into Left ($0 - 33\%$), Center ($33 - 67\%$), and Right ($67 - 100\%$), the system translates coordinates into natural directional guidance (*"Steer Left"*, *"Walk Straight"*, *"Steer Right"*), matching how a human guide communicates.

Q9: How does the system handle high-latency video streams from mobile IP cameras over Wi-Fi?

Architectural Answer:
Network streams experience variable packet delivery. Standard OpenCV buffers frames, creating a 1–3 second visual delay. Our `ThreadedCameraStream` runs an independent capture loop with `CAP_PROP_BUFFERSIZE = 1`. New frames overwrite the internal buffer, immediately discarding stale frames so the inference pipeline always processes the latest live view.

Q10: How does the Text Reading subsystem maintain reading order across multi-column documents?

Architectural Answer:
OCR engines output bounding boxes in arbitrary detection order. Feature 6 applies a 2D spatial reading-order sort:

$$\text{Sort Key} = \left(\text{round}\left(\frac{y_{\text{center}}}{30}\right), x_{\text{center}}\right)$$

This groups words into horizontal line bands (within 30px vertical tolerance) and sorts each band left-to-right, ensuring natural reading flow across sentences and paragraphs.

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10. Conclusion, Deployment Roadmap & Compilation Guide

10.1 Architectural Achievements

The AI Assistive Vision System successfully integrates deep learning, monocular optical geometry, classical spatial computer vision, and hands-free voice orchestration into a unified platform. Operating at 28–60 FPS on standard edge CPU hardware, it provides a comprehensive mobility and cognitive assistance solution for visually impaired users.

10.2 Future Production Roadmap

1. **Edge TPU / NPU Porting**: Exporting MobileNetV3 and YOLOv8 to TensorRT and INT8 ONNX for deployment on wearable compute units (e.g. Raspberry Pi 5 with AI Kit / Jetson Orin Nano).
2. **Haptic Belt Integration**: Complementing audio voice cues with directional vibrating motors for noisy outdoor environments.
3. **GPS & OpenStreetMap Fusion**: Merging micro-obstacle avoidance with macro-turn-by-turn outdoor street navigation.

10.3 Professional PDF Compilation Instructions

This report is written in GitHub Flavored Markdown with embedded high-resolution figures, LaTeX equations, and Mermaid diagrams. It can be compiled into a PDF using any of the following methods:

Method A: VS Code Markdown PDF Extension (Recommended)

1. Open this file in VS Code.
2. Install the **Markdown PDF** (`yzane.markdown-pdf`) extension.
3. Right-click anywhere in the editor and select **Markdown PDF: Export (pdf)**.
4. The generated PDF will retain all tables, formulas, and embedded graphics.

Method B: Pandoc with wkhtmltopdf / Weasyprint

```
pandoc comprehensive_architectural_report.md -o AI_Eye_Architectural_Report.pdf --pdf-engine=weasyprint
```

Method C: Chrome Headless Print

```
# Convert to HTML then print to PDF via Chrome Headless:
google-chrome --headless --print-to-pdf=AI_Eye_Architectural_Report.pdf comprehensive_architectural_report.m
```

End of Engineering Specification Report.